

Title: Transient cell states encode positional information to direct asymmetric growth

Authors: Travis A. Lee^{1,2}, Kamonkan Gamnerdsiri¹, Natanella Illouz-Eliaz¹†, Tatsuya Nobori¹‡, Joseph R. Nery³, Bruce Jow³, Michelle Liem⁴, Caz O'Connor⁴, Joseph R. Ecker^{2,3*}

Affiliations:

¹Plant Biology Laboratory, The Salk Institute for Biological Studies, La Jolla, CA, USA.

²Howard Hughes Medical Institute, The Salk Institute for Biological Studies, La Jolla, CA, USA

³Genomic Analysis Laboratory, The Salk Institute for Biological Studies, La Jolla, CA, USA.

⁴Flow Cytometry Core Facility, The Salk Institute for Biological Studies, La Jolla, CA, USA

*Corresponding author. Email: ecker@salk.edu

†Present address: The Sainsbury Laboratory, University of East Anglia, Norwich, UK

Abstract: How tissues generate asymmetric growth from spatially and temporally restricted signals remains a fundamental challenge in developmental biology. We show that transient cell states encode positional information that instructs asymmetric growth in plants. Using spatial single-cell transcriptomics and multi-omics, we uncover a surprisingly complex landscape of transient cell states in the apical hook, including a previously unrecognized population in the apical hypocotyl region that integrates developmental and hormonal cues and bifurcates into opposing growth trajectories. Gene regulatory network analysis and functional perturbation establish GATA TRANSCRIPTION FACTOR 2 (GATA2) as a central regulator that promotes cell elongation through gibberellin signaling. These results reveal that diverse, spatially localized transient cell states function as regulatory hubs that convert positional information into divergent growth programs, providing a unifying framework for how dynamic cellular states shape organ morphology.